

$$p \rightarrow q \equiv \neg p \vee q$$

$$\neg p \rightarrow q \equiv p \vee q$$

$$p \wedge q \equiv \neg(\neg p \vee \neg q) \equiv \neg(p \rightarrow \neg q)$$

Ex.:

$$p \rightarrow q, r \rightarrow p, r \quad \vdash^? q$$

$$f_1: p \rightarrow q$$

$$f_2: r \rightarrow p$$

$$f_3: r$$

$$f_2, f_3 \vdash_{mp} f_4: p$$

$$f_1, f_4 \vdash_{mp} f_5: q$$

$f_1, f_2, \dots, f_5$ este deducția lui q din ipoteze

$$\text{Deci } p \rightarrow q, r \rightarrow p, r \vdash q$$

$\vdash$ teoremă

$$\vdash^? u \rightarrow u$$

$$f_1: (u \rightarrow ((u \rightarrow u) \rightarrow u)) \rightarrow ((u \rightarrow (u \rightarrow u)) \rightarrow (u \rightarrow u)) \quad A_2$$

$$f_2: u \rightarrow ((u \rightarrow u) \rightarrow u) \quad A_1$$

$$f_1, f_2 \vdash_{mp} f_3: (u \rightarrow (u \rightarrow u)) \rightarrow (u \rightarrow u)$$

$$f_4: u \rightarrow (u \rightarrow u) \quad A_1$$

$$f_3, f_4 \vdash_{mp} f_5: u \rightarrow u$$

Deci $f_1, \dots, f_5$ este deducția lui $u \rightarrow u$, deci $\vdash u \rightarrow u$

$\vdash^? u \wedge v \rightarrow v$
 $\vdash^? \neg(u \rightarrow \neg v) \rightarrow v$
 $f_1: (\neg v \rightarrow (u \rightarrow \neg v)) \rightarrow (\neg(u \rightarrow \neg v) \rightarrow v) \quad A_3$
 $f_2: \neg v \rightarrow (u \rightarrow \neg v) \quad A_1$
 $f_1, f_2 \vdash_{\text{imp}} f_3: \neg(u \rightarrow \neg v) \rightarrow v$
 f_1, f_2, f_3 e deductia lui $u \wedge v \rightarrow v$, deci $\vdash u \wedge v \rightarrow v$

$\vdash^? u \rightarrow ((u \rightarrow v) \rightarrow v)$
 $\xrightarrow{i\Delta} u \vdash^? (u \rightarrow v) \rightarrow v$
 $\xrightarrow{i\Delta} u, u \rightarrow v \vdash^? v$

$f_1: u \quad \text{ip.}$
 $f_2: u \rightarrow v \quad \text{ip.}$
 $f_1, f_2 \vdash_{\text{imp}} f_3: v$
 Deci f_1, f_2, f_3 este deductia lui v din u si $u \rightarrow v$, deci $u, u \rightarrow v \vdash v$

$\xrightarrow{T\Delta} u \rightarrow v \vdash u \rightarrow v$
 $\xrightarrow{T\Delta} u \rightarrow (u \rightarrow v) \rightarrow v$

$u, u \rightarrow v \vdash v$
 $\xrightarrow{T\Delta} u \rightarrow (u \rightarrow v) \rightarrow v$
 $\xrightarrow{T\Delta} \vdash u \rightarrow ((u \rightarrow v) \rightarrow v)$

Ex. partial:

$$\begin{array}{r}
 1) \quad 1110101101111000 + \\
 \quad 1000101110101001 \\
 \hline
 10111011100100001
 \end{array}$$

$$\begin{array}{r}
 2) \quad 512341(6) - \\
 \quad 401253(6) \\
 \hline
 111044(6)
 \end{array}$$

$$3) \overset{+2}{A} \overset{+2}{1B} \overset{+2}{2C} 3D (16) \cdot$$

$$\quad \quad \quad \underline{3(16)}$$

$$1E5184B7(16)$$

$$\begin{array}{r} 13. \\ \underline{3} \\ 39 \end{array}$$

$$39:16=2$$

$$\begin{array}{r} 32 \\ \underline{=7} \end{array}$$

$$\begin{array}{r} 12. \\ \underline{3} \\ 36 \end{array}$$

$$\begin{array}{r} 11. \\ \underline{3} \\ 33 \end{array}$$

$$30:16=1 \text{ rest } 14$$

Virgulă fixă

$$8904,37(10)=$$

$$8904,37(10)=20310,2$$

$$8904:8=1113=139=16=2=0$$

$$01302$$

$$\begin{array}{r} 0,37. \\ \underline{8} \\ 2,96. \\ \underline{8} \end{array}$$